ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

ENGINEERING MATHEMATICS II

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 2nd year final examination: first-order and second-order differential equations, the Laplace transform, multivariable calculus (partial and directional derivatives, extrema, multiple integrals, line integrals and Green's theorem), eigenvalues, series and Fourier series.

Written by Academic Central for practice, following second-year engineering mathematics modules at South African universities (e.g. University of Pretoria WTW 256 Differential Equations, WTW 258 Calculus and WTW 238 Mathematics). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working. Marks are given for correct steps, not only for final answers.
3. A table of Laplace transforms is NOT provided; standard transforms must be known.
4. Non-programmable scientific calculators may be used. Give answers correct to TWO decimal places unless stated otherwise.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: FIRST-ORDER DIFFERENTIAL EQUATIONS

1.1 Solve the separable equation $\frac{dy}{dx} = xy^2$, given $y(0) = 1$. (5)

1.2 Use an integrating factor to solve $\frac{dy}{dx} + 2y = 4$, given $y(0) = 0$. (5)

1.3 A motor housing cools according to Newton's law of cooling, $\frac{dT}{dt} = -k(T - 20)$, where T is in °C and t in minutes. Its temperature is 90 °C at $t = 0$ and 60 °C at $t = 10$. Find k, and find the time at which the temperature reaches 30 °C. (5)

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QUESTION 2: SECOND-ORDER LINEAR DIFFERENTIAL EQUATIONS

2.1 Find the general solution of $y'' - 5y' + 6y = 0$. (3)

2.2 Solve $y'' + 4y' + 4y = 0$, given $y(0) = 1$ and $y'(0) = 0$. (4)

2.3 A mass-spring-damper system is modelled by $y'' + 2y' + 5y = 0$. Find the general solution and state whether the system is underdamped, critically damped or overdamped. (4)

2.4 Use the method of undetermined coefficients to find the general solution of $y'' - y = 3e^{2x}$. (5)

2.5 Write down the FORM of a particular solution of $y'' + 9y = 2\sin 3x$ (do not find the constants), and explain why the usual form must be changed. What happens physically to a system like this? (4)

[20]

QUESTION 3: THE LAPLACE TRANSFORM

3.1 Use the definition of the Laplace transform to show that $\mathcal{L}\{e^{at}\} = 1/(s - a)$ for $s > a$. (3)

3.2 Find $\mathcal{L}\{3t^2 - 2\sin 4t + e^{-t}\}$. (3)

3.3 Find the inverse Laplace transform of $F(s) = (s + 5) / ((s + 1)(s + 2))$. (4)

3.4 Use the Laplace transform to solve $y' + 2y = 4$, given $y(0) = 1$. (5)

3.5 Use the Laplace transform to solve $y'' + 3y' + 2y = 0$, given $y(0) = 0$ and $y'(0) = 1$. (5)

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QUESTION 4: MULTIVARIABLE CALCULUS

4.1 Given $f(x, y) = x^2y + 3xy^2$, find $\partial f/\partial x$, $\partial f/\partial y$ and $\partial^2 f/\partial x \partial y$. (3)

4.2 For the function in 4.1, find the directional derivative at the point (1; 2) in the direction of the vector $3i + 4j$. What is the largest rate of change of f at that point? (4)

4.3 Find and classify all the stationary points of $f(x, y) = x^3 - 3x + y^2$. (5)

4.4 Evaluate the double integral $\int$ from $x = 0$ to 1 $\int$ from $y = 0$ to 2 of $(x + 2y)\, dy\, dx$. (3)

4.5 Use polar coordinates to evaluate $\iint e^{-(x^2 + y^2)}\, dA$ over the disc $x^2 + y^2 \leq 1$. (4)

4.6 Show that the vector field $F = y\, i + x\, j$ is conservative, and use this to evaluate the line integral of F along the curve $y = x^2$ from (0; 0) to (1; 1). (3)

4.7 Use Green's theorem to evaluate $\oint (-y\, dx + x\, dy)$ anticlockwise around the circle $x^2 + y^2 = 1$. (3)

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QUESTION 5: EIGENVALUES AND SERIES

5.1 Find the eigenvalues and corresponding eigenvectors of the matrix $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. (6)

5.2 Hence write down the general solution of the system of differential equations $x' = Ax$, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. (2)

5.3 Use the ratio test to decide whether the series Σ ($n = 1$ to ∞) of $n / 3^n$ converges or diverges. (3)

5.4 A square wave has period 2π and is defined by $f(x) = -1$ for $-\pi < x < 0$ and $f(x) = 1$ for $0 < x < \pi$. Find its Fourier series. (6)

5.5 Find the radius of convergence of the power series Σ ($n = 1$ to ∞) of x^n / n. (3)

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TOTAL: 100

END OF QUESTION PAPER